

ORIGINAL ARTICLE

Weekly paclitaxel for platin-resistant stage IV head and neck cancer patients

JUAN JOSÉ GRAU¹, MIGUEL CABALLERO², EUGENIA VERGER³,
MARIANO MONZÓ⁴ & JOSE LUÍS BLANCH²

¹Oncology Department, ICMHO (Institut Clínic de Malalties Hemato-Oncològiques), ²Otorhinolaryngology Department, ³Radiation Oncology Department and ⁴Human Anatomy Department, University of Barcelona, IDIBAPS (Augusto Pi i Suñer Memorial Institute for Biomedical Research), Hospital Clínic of Barcelona, Barcelona, Spain

Abstract

Conclusions. Weekly paclitaxel may be an active and well tolerated chemotherapy regimen for patients with platin-resistant advanced head and neck cancer. **Objectives.** Weekly paclitaxel should be an active and well tolerated regimen for palliative treatment of platin-resistant patients with recurrent or metastatic carcinoma of the head and neck. We analyzed the antitumor activity and toxicity profile. **Patients and methods.** Sixty consecutive patients with advanced head and neck cancer were treated with 1 h infusion of paclitaxel, 80 mg/m² weekly, for 6 consecutive weeks. Patients who showed disease response or disease stabilization continued until progression of disease. **Results.** A total of 719 doses of paclitaxel were administered to the 60 patients. No complete response was observed. Partial response and stable disease were observed in 26 (43.3%) and 9 (15%) patients, respectively. Median time to tumor progression for patients who responded to therapy was 6.2 months (SD = 1.3; 95% CI, 3.7–8.6) and the overall median survival in this group of patients was 8.5 months (SD = 1.4; 95% CI, 5.7–11.2). The main toxic effects were leukopenia (26.6%), anemia (43.3%), fatigue (37.4%), alopecia (18.7%), rash/desquamation (13.3%), and thrombophlebitis (6.8%).

Keywords: Advanced tumors, head and neck cancer, squamous cell carcinoma, chemotherapy

Introduction

Patients with advanced head and neck cancer resistant to platinum-based chemotherapy in whom no other local therapies can be administered have a very poor prognosis. Chemotherapy in these patients induces a low response rate and short survival. Previously, in these patients treated with different therapies a 3.4% response rate with an overall survival of 3.4 months (95% CI, 2.6–4.2) has been reported [1]. An additional problem is the poor physical condition of these patients who have previously been heavily pretreated, as they may have a poor tolerance to standard doses of chemotherapy. In these patients metronomic chemotherapy (consisting of low dose with increased frequency) can be an option. Metronomic chemotherapy induces high rates of response, of long duration, reducing acute toxicity in some tumors [2]. From previous preclinical studies, continuous low

dose metronomic chemotherapy is thought to inhibit tumor angiogenesis. This suggests that activated endothelial cells may be more sensitive to chemotherapeutic drugs than tumor cells [3,4].

The potential antiangiogenic effect of an active metronomic chemotherapy could be confirmed in vivo, including patients with head and neck cancer mainly in advanced stages [5,6]. There is a strong correlation of angiogenesis with tumor stage and with the outcome in head and neck cancer [7]. An antiangiogenic therapy should be an interesting treatment to be tested in these patients. Paclitaxel has been involved in the inhibition of cancer angiogenesis [8]. It is common clinical practice in many countries to use a low dose of weekly paclitaxel. There are many studies of paclitaxel associated with other drugs or radiotherapy, particularly in the neoadjuvant setting [9]. In patients

Correspondence: Miguel Caballero PhD, Otorhinolaryngology Department, Hospital Clínic of Barcelona, University of Barcelona, Villarroel St 170 – 08036, Barcelona, Spain. Tel +34 93 227 5400. Fax +3493 2275454. E-mail: mcaba@clinic.ub.es

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with advanced head and neck cancer, a low dose of weekly paclitaxel has been tested in a phase II study that included 48 patients. In that study (published in abstract form) a 9.3% response rate with 44% tumor stabilizations was observed [10]. The median duration of response was 3.2 months and the main toxicity was neutropenia or neuropathy grade 3 in 8% and 4% of the patients, respectively. Nevertheless, there is no study published as a full paper analyzing the efficacy and toxicity of weekly paclitaxel alone in these patients. In this study we prospectively analyzed the clinical outcome and feasibility of low dose weekly paclitaxel in patients with advanced head and neck cancer.

Patients and methods

Patient population

Patients with recurrent and/or metastases from advanced head and neck cancer with a confirmed diagnosis of disease progression to platinum-based chemotherapy were included in the study. Documented tumor progression on prior systemic induction chemotherapy plus chemoradiotherapy before 1 year of the end of treatment or no more than one prior systemic palliative chemotherapy was allowed. Patients had to be completely recovered from the acute toxic effects of prior first-line treatment. Additional eligibility criteria included unidimensionally measured disease according to Response Evaluation Criteria in Solid Tumors (RECIST) [11], an Eastern Cooperative Oncology Group (ECOG) performance status (PS) of two or less, a life expectancy of at least 2 months, and adequate bone marrow, hepatic, and renal function. Patients with documented brain metastases, clinically significant third-space fluid collections (e.g. ascites or pleural effusions) or second primary malignancies treated in the last 5 years of entry were excluded. Pregnant or breast-feeding women and patients unable to use an approved contraceptive method during and for 3 months after the study were also excluded. The study was approved by the ethical committee of our institution and each patient gave written informed consent before the study inclusion. This study was conducted according to the Helsinki Declaration and according to the guidelines of good clinical practice.

Treatment plan and dose modifications

The patients who were enrolled received 80 mg/m² of paclitaxel via a 1 h infusion schedule administered weekly. Premedication consisted of 8 mg of dexamethasone i.v., 50 mg of diphenhydramine i.v., and 50 mg of ranitidine i.v. Treatment was administered

for 6 consecutive weeks on an outpatient basis. Before every dose of paclitaxel, a complete blood examination, anamnesis, and physical examination for toxicity was carried out. In the case of clinical response or stabilization, patients continued with weekly paclitaxel for 6 months or until objective progression of disease, intolerance, toxicity, or voluntary patient withdrawal.

When acute toxicity appeared the treatment was delayed 1 week. When toxicity persisted continuously for 3 weeks, the treatment was discontinued.

Response assessment

Response evaluation criteria were used according to the standard RECIST [11]. CT or MRI scan, chest radiographs, and upper respiratory tract exam were used as baseline and every six doses or when disease progression was suspected. Response was first evaluated after six doses and confirmatory scans were obtained 4–6 weeks after initial documentation of an objective response. Patients were examined by a physician every week before treatment.

Evaluations were performed according to the intention to treat. All patients who had received at least two doses of paclitaxel were included in the analyses of time to progression, overall survival, and toxicity.

Statistical analysis

The primary outcome measure of this study was the 3-month survival rate. Overall survival was defined as the time from the first treatment administration to the date of death from any cause. Assuming a 3-month survival rate or 50%, no censoring, and a sample size of 60 patients, the width of corresponding 95% confidence interval (CI) was regarded as having an adequate precision. The time to progression (TTP) was defined as the time from treatment administration to the first observation of disease progression or death. The TTP was censored at the date of death for patients who died without documented disease progression or at the date of the last visit for those patients who were alive without documented disease progression. All time to event parameters were analyzed according to the Kaplan and Meier method [12]. All further data for categorical or continuous variables are given as absolute and relative frequencies or as medians with ranges, respectively, if not otherwise indicated. The variables that were significant in the univariate analysis were introduced into a logistical regression model for the multivariate analysis (back step Wald's method) to determine independent risk factors for response, using an input variable criterion ($p < 0.05$) and an

output variable criterion ($p > 0.10$). Both univariate and multivariate Cox regression were used to analyze factors potentially affecting overall survival. Statistical significance in this study was set as $p < 0.05$. All analyses were performed with SPSS statistical package version 13.0 (SPSS Inc., Chicago, IL, USA).

Results

Patient characteristics

From October 2001 to December 2006, 60 consecutive patients were enrolled in the study, with a follow-up for survival until May 2007. All the patients received at least two doses of paclitaxel and were, therefore, assessable for survival, response, and toxicity. Patient disease characteristics are summarized in Table I. The median age at the start of chemotherapy was 59.5 years (range 45–79 years). The predominant site of the disease was local or regional relapse in 31 (51.7%) patients, metastases in 13 (21.7%), and both in 6 other patients (26.4%). The primary tumor was located in the oropharynx in

Table I. Characteristics of patients.

Characteristics	n (%)
Total	60
Gender	
Male	55 (91.7)
Female	5 (8.3)
Age (years)	
Median (SD)	59.5 (8.3)
Range	45–79
ECOG performance status	
0	1 (1.7)
1	50 (83.3)
2	9 (15.0)
Location of primary tumor	
Larynx	20 (33.3)
Oropharynx	30 (50)
Hypopharynx	10 (16.7)
Disease status	
Local recurrence	31 (51.7)
Metastases	13 (21.7)
Both	16 (26.4)
Previous cancer treatment	
Surgery	38 (62.3)
Radiotherapy alone	15 (24.6)
Chemotherapy plus radiotherapy	7 (11.5)
Previous palliative chemotherapy	
No	28 (46.7)
Yes	32 (53.3)

ECOG, Eastern Cooperative Oncology Group.

30 patients (50%), hypopharynx in 10 patients (16.7%), supraglottis in 13 patients (21.7%), or glottis in 7 patients (11.7%). Thirty-two patients (53.3%) received one line of palliative chemotherapy before weekly paclitaxel. In 28 other patients, paclitaxel was given as a first line of palliative chemotherapy because of early relapse after surgery plus adjuvant chemoradiotherapy (9 patients) or after primary treatment with chemoradiotherapy for non-operable tumors (16 patients) or for conservative treatment (3 patients).

Treatment

A total of 719 doses of paclitaxel were administered in this study, with a median of 12 doses per patient (range 2–31). The median accumulated dose of paclitaxel per patient was 960 mg/m² (range 160–2480 mg/m²). At the time of the final analysis all the 60 patients had discontinued the study treatment. The main reasons for discontinuation were disease progression (40 patients, 67%), adverse events (8 patients, 13%), death from the study disease (6 patients, 10%), end of programmed treatment (4 patients, 7%), and patient refusal (2 patients, 3%). No dose reduction was performed. When hematological or other acute toxicity appeared, paclitaxel was delayed from 1 to 3 weeks.

Response

Tumour response is summarized in Table II. No complete response was observed by imaging criteria. Overall, 26 (43.3%) patients achieved a partial response (PR) (30% decrease in the sum of the longest diameter of target lesions and no appearance of new lesions), stable disease was observed in 9 (15%) patients, and clinical benefit (partial response plus stable disease) in 34 (58.3%) of patients. The median time to progression for responding patients was 6.2 months (SD = 1.3; 95% CI, 3.7–8.6). Logistical regression did not show that age, gender,

Table II. Weekly paclitaxel activity.

Parameter	n (%)
Complete response (CR)	0 (0)
Partial response (PR)	26 (43.3)
Stable disease (SD)	9 (15.0)
Clinical benefit (CR+PR+SD)	35 (58.3)
Progressive disease	23 (38.3)
Non-evaluable	2 (3.3)

primary location of the tumor, performance status or stage (IVb vs VIc) were independent predictive factors for response.

Survival

Fifty-two (86.7%) patients had died and 8 (13.3%) other were still alive. The 3-month survival rate was 80%. The median overall survival time was 8.5 months (SD = 1.4; 95% CI, 5.7–11.2) for responding patients, 3.4 months (SD = 0.7; 95% CI, 2.0–4.9) for non-responding patients, and 5.2 (SD = 0.4; 95% CI, 4.5–6.0) months for the global group of patients (Figure 1). Univariate and multivariate analysis by Cox regression showed that response to chemotherapy (OR 0.43; 95% CI 0.24–0.77; $p = 0.004$) was the only prognostic factor in the overall survival. Age, gender, primary location, performance status or stage (IVb vs VIc) was independently associated with the survival.

Toxicity

Toxicity assessment of the 60 patients is detailed in Table III. Hematologic toxicities observed were leukopenia (26.7%), anemia (43.3%), and thrombophlebitis (6.7%). No platelet toxicity was recorded. The nonhematologic toxicities observed were diarrhea (5.0%), nausea (5.0%), fatigue (36.7%), alopecia (18.3%), paresthesia (5.0%), rash/desquamation (13.3%), and stomatitis/pharyngitis (10.0%). There were no episodes of fluid retention. No patients discontinued therapy because of toxicity. There were no treatment-related deaths during the study.

Discussion

This study on 60 patients is, as far as we know, the largest completed phase II study to evaluate the role of weekly paclitaxel in patients with platin-resistant

advanced head and neck cancer. We have observed a 43.3% response rate with weekly low dose of paclitaxel in these patients. The overall survival in the responding group has a median of 8.5 months, and the toxicity was rather low. These data are clearly better than those reported by other authors in a similar set of patients, receiving best supportive care or other second-line chemotherapies [1]. These results have recently been improved by a cooperative group (in which we also participate) adding the anti-epidermal growth factor monoclonal antibody, cetuximab, to chemotherapy [13]. In a phase II study of 46 patients, with similar inclusion criteria, weekly paclitaxel (80 mg/m²) was added to cetuximab, 400 mg/m², followed by weekly infusion of 250 mg/m². Complete response was observed in 20% of the patients, and partial response in another 51%. Until now, chemotherapy alone has shown poor results in this set of patients. Nevertheless, the inclusion of low and weekly doses of taxoids as palliative treatment for advanced head and neck cancer provided hopeful results. Hitt et al. [14] reported a 42% response rate and an estimated median overall survival of 11.3 months in a series of 38 patients treated with weekly low dose of docetaxel as a first-line chemotherapy for patients with advanced head and neck cancer, with an excellent toxicity profile. Both platin-resistant and naive patients were included. These results were similar to those obtained by us, but we included previously treated platin-resistant patients only.

The purpose of chemotherapy in these patients is to obtain symptom control, prevent serious complications, and increase survival with improvement of quality of life. It is very important to have a treatment for such patients that allows good tolerance and low toxicity. Weekly low dose of paclitaxel offers low acute toxicity, particularly hematologic toxicity. We observed only a few patients with clinical grade 3 paresthesia that did not decrease the performance status and appeared after many weeks of therapy. The major toxicity was chronic anemia, which could be improved with the concomitant administration of oral iron support and, in some cases, the administration of subcutaneous erythropoietin analogs.

How the chronic administration of low doses of chemotherapy can improve the therapeutic index over the classic maximum tolerated doses is a question that could be explained by many reasons. The good tolerability may allow a better compliance by the patients. As a consequence, low doses of chemotherapy became a dose-dense therapy that can explain improved therapeutic results. Nevertheless, sometimes, total dose of metronomic chemotherapy might be significantly less than maximum tolerated

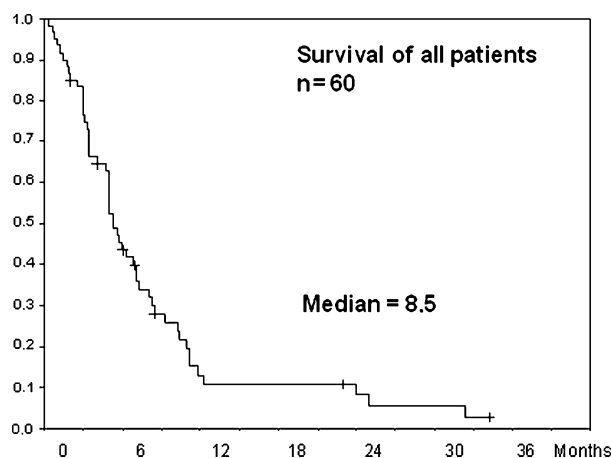


Figure 1. Overall survival.

Table III. Toxicity of weekly paclitaxel.

Toxicity	No. of patients (%) by toxicity grade			
	Grade 1	Grade 2	Grade 3	Grade 4
Leukopenia	11 (18.3)	3 (5.0)	1 (1.7)	1 (1.7)
Thrombocytopenia	–	–	–	–
Anemia	17 (28.3)	8 (13.3)	1 (1.7)	–
Thrombophlebitis	–	1 (1.7)	3 (5.0)	–
Diarrhea	–	3 (5.0)	–	–
Nausea	–	3 (5.0)	–	–
Paresthesia	–	–	3 (5.0)	–
Fatigue	17 (28.3)	5 (8.3)	–	–
Alopecia	11 (18.3)	–	–	–
Rash/desquamation	8 (13.3)	–	–	–
Stomatitis/pharyngitis	5 (8.3)	1 (1.7)	–	–

dose achieving similar or even superior therapeutic results. Maximum tolerated dose chemotherapy mainly targets proliferating tumor cells, while low dose chemotherapy appears to inhibit preferentially the endothelial cells of the tumor growing vasculature [15]. Paclitaxel, in very low doses, selectively inhibits the proliferation of human endothelial cells but does not affect the cellular microtubule structure, and the treated cells do not show G2/M cell cycle arrest and apoptosis, suggesting a novel mechanism of action [16]. Activated endothelial cells may be more sensitive, or even selectively sensitive, to protracted ('high-time') low dose chemotherapy compared with other types of normal cells, resulting in apoptosis of endothelial cells enhancing the therapeutic effect [17]. We observed a rate of 6.8% of thrombophlebitis as a toxic effect that can be related to the endothelial therapeutic activity of weekly paclitaxel.

Maximum tolerated doses of paclitaxel have also been chosen in these patients. A 40% response rate has been reported with paclitaxel at 250 mg/m² every 3 weeks in 23 patients with advanced head and neck cancer [18]. In this study, severe or life-threatening granulocytopenia was the most frequent toxicity observed, occurring in 91% of patients. The comparison between the results of phase II studies suggests that a low dose of chemotherapy can be better than higher doses of the same drug in this set of patients. These results may open the way to trials with other new antiangiogenic agents such as erlotinib, bevacizumab or others, which could induce tumor response in patients with advanced head and neck cancer. On the other hand, weekly paclitaxel is an active and well tolerated chemotherapy that can be combined with other new agents such as cetuximab as palliative treatment for advanced head and neck cancer patients, to improve the previous poor results obtained until now in these patients.

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